

SMART WATER PUMP AUTOMATION SYSTEM

ESP32 + Blynk IoT | Automated 2 HP & 6 HP Pump Control

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01 | PROJECT OVERVIEW

From sensor inputs to cloud-based motor supervision.

SMART CONTROL

Automates and remotely monitors 2 HP and 6 HP water pumps.

ESP32 CORE

Control, sensing, timing and protection logic.

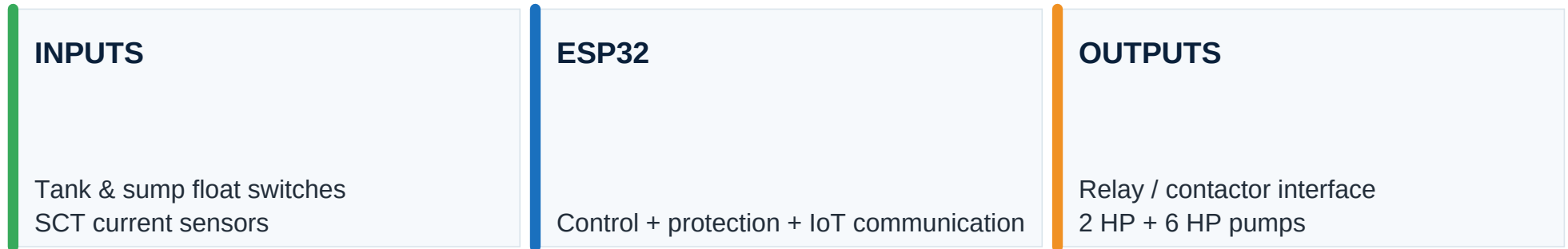
BLYNK IoT

Live status, current, runtime, mode and fault information.

Key capabilities: Automatic / Manual operation • Tank & sump supervision • Over-current and dry-run protection • Runtime and delay management • Remote monitoring.

02 | SYSTEM ARCHITECTURE

Blynk Cloud ↔ Wi-Fi ↔ ESP32 ↔ Sensors / Protection ↔ Relay-Contactor ↔ Pumps.



Software stack: Arduino IDE • Embedded C++ • ESP32 Arduino core • Blynk IoT • EmonLib • BlynkTimer

03 | MOTOR AUTOMATION LOGIC

Automatic operating sequence for both pumps.

2 HP — AUTO

TANK LOW → START 2 HP → MONITOR → TANK FULL → STOP

Trips: Over Current • Dry Run • Sump Dry • Maximum Runtime

6 HP — AUTO

SUMP DRY + NOT FULL → START 6 HP → MONITOR → SUMP FULL → STOP

Trips: Over Current • Dry Run • Maximum Runtime

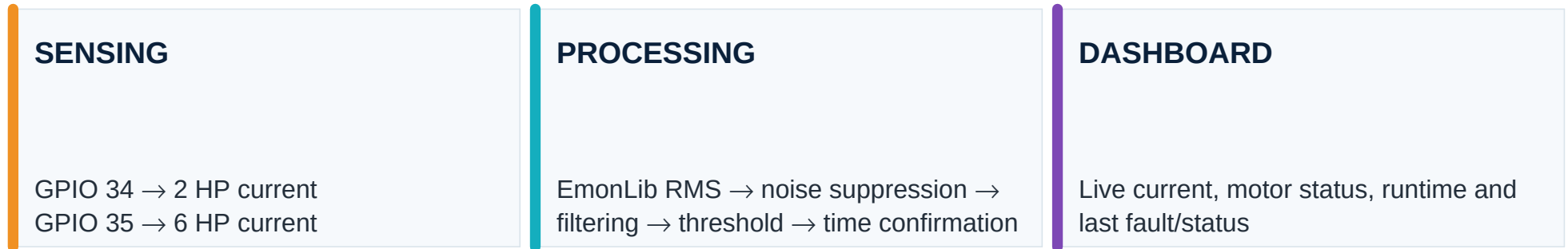
04 | PROTECTION & TIMING

Configured firmware values; verify against actual motor and field protection requirements.

Parameter	Value	Purpose
Over-current	13.0 A	Configured threshold
Dry-run	1.0 A	Current-based detection
Startup delay	10 s	Protection delay
OC confirmation	3 s	Condition must persist
Dry-run confirmation	10 s	Condition must persist
Restart delay	5 s	Prevents immediate restart
Normal runtime	15 min	Normal setting
Test runtime	50 s	Bench testing

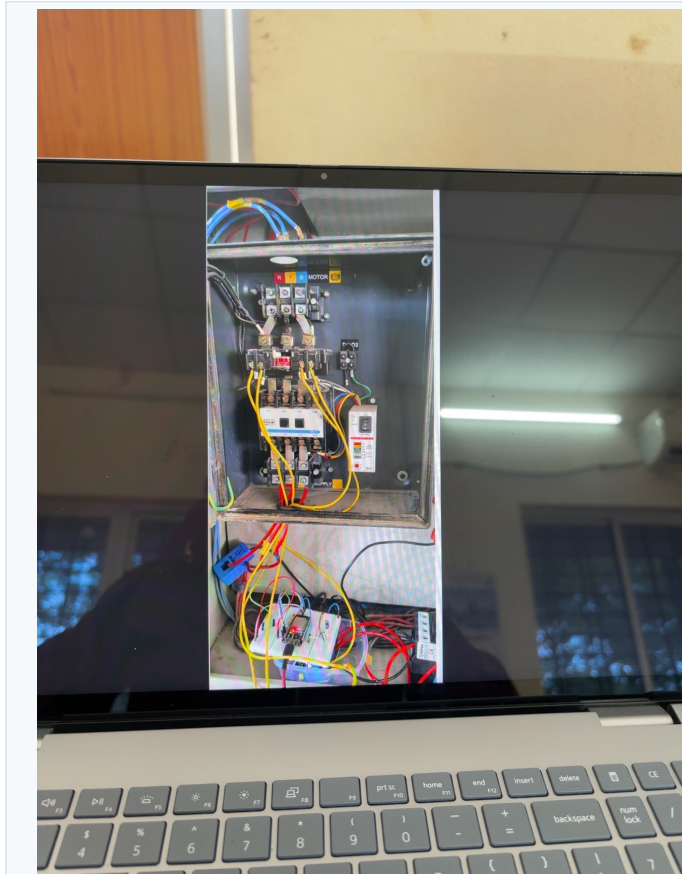
05 | CURRENT MONITORING

RMS current measurement with filtering and protection confirmation.



06 | HARDWARE IMPLEMENTATION

Actual motor control panel and ESP32 prototype.



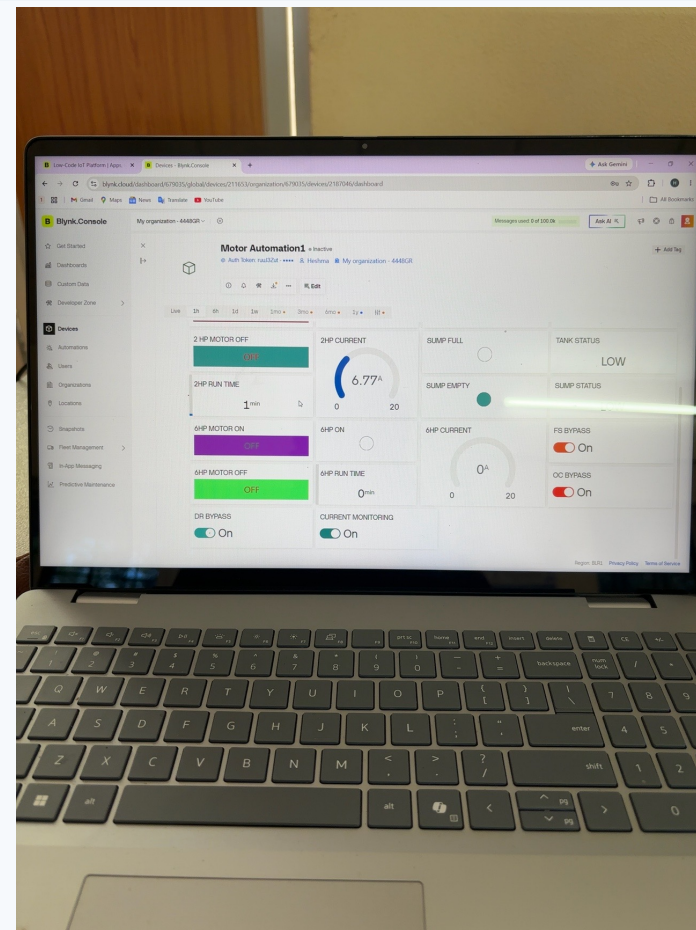
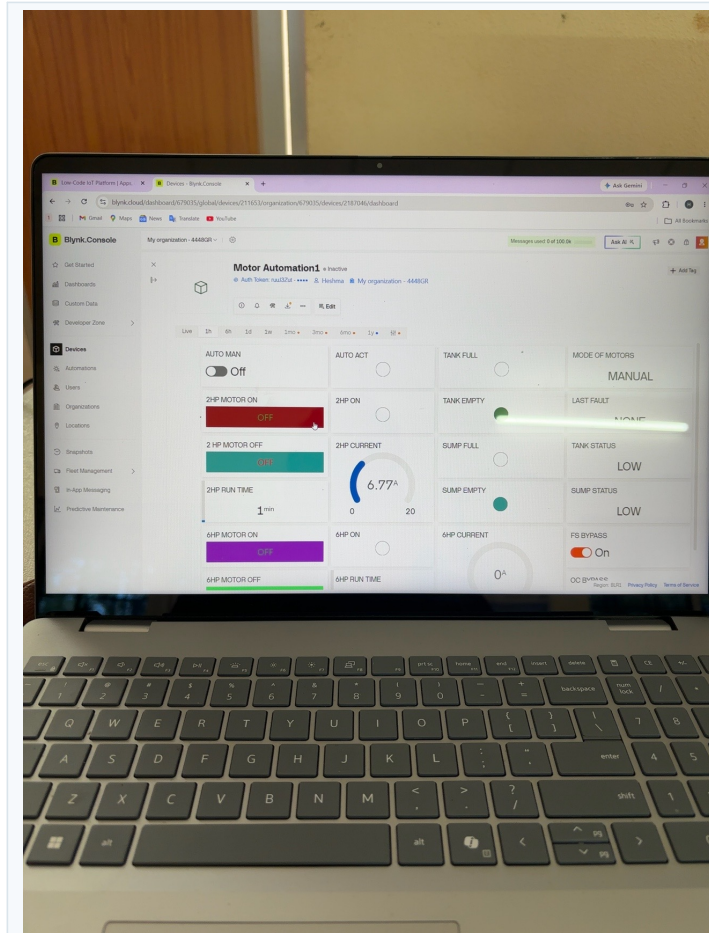
Prototype elements

- Three-phase motor/control section
- Contactor and control wiring
- Low-voltage relay/control interface
- Current-sensing interface
- ESP32 prototype wiring

Motor power circuits require correctly rated protection, isolation and earthing.

07 | BLYNK IoT DASHBOARD

Actual project dashboard screenshots.



08 | DASHBOARD FUNCTIONS & RESULT

AUTO / MANUAL

Operating mode selection.

2 HP / 6 HP

Separate start/stop control.

CURRENT

Live current gauges.

TANK / SUMP

Level-condition indication.

RUNTIME / FAULT

Runtime and last-fault/status.

BYPASS / CLOUD

Controlled service bypasses and Blynk monitoring.

The ESP32 firmware integrates motor control, water-level sensing, current measurement and protection logic.

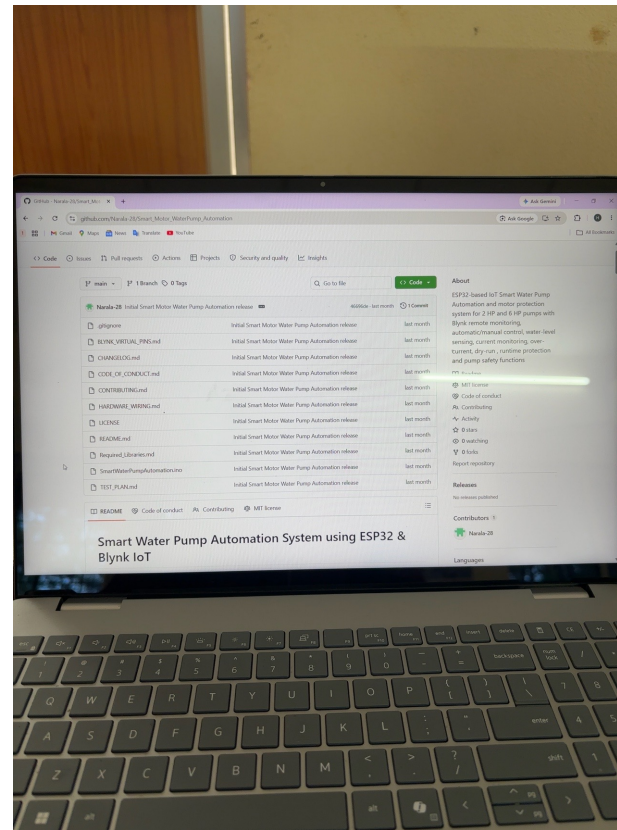
09 | DEVELOPER & GITHUB

The project is documented as a complete engineering repository.

DEVELOPER Sree

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Embedded C++ • ESP32 • IoT/Blynk • Automation
• Motor protection



10 | FUTURE SCOPE

NEXT DEVELOPMENT

Energy measurement • Power factor • Flow sensor • Water consumption logging • Web dashboard • MQTT • OTA updates • GSM/4G backup • Historical cloud data • Advanced alarm management • Improved PCB protection/isolation

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ESP32 • Embedded C++ • Blynk IoT • Protection • Automation